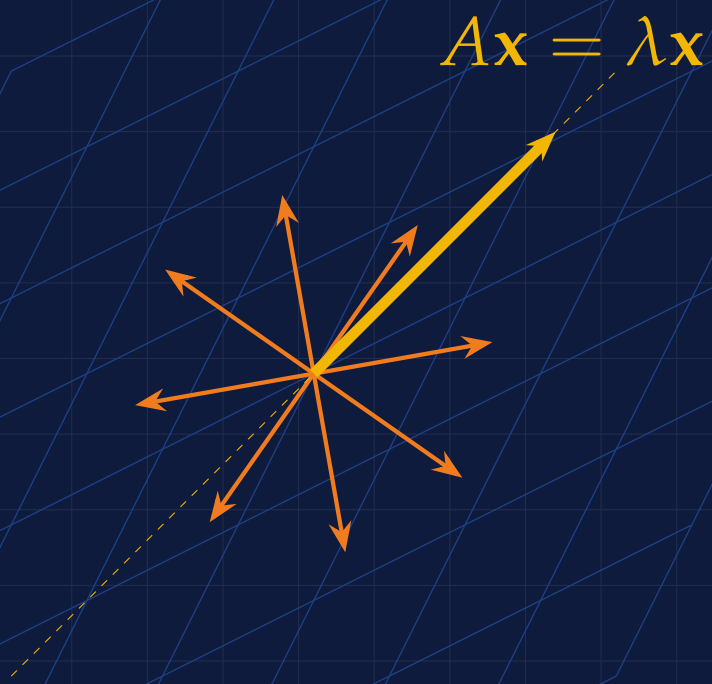
ZMathLab • Linear Algebra

Eigenvalues, Eigenvectors & Diagonalization

The arrows a matrix cannot turn: how to find them, why they matter, and how they turn a hard matrix into a diagonal one.



Muhammad Zubair

BS Linear Algebra • Notes with solved examples

Based on Anton & Rorres §5.1--5.2 and Williams §3.4

How to use these notes

Every idea comes in the same order: a **story** (why we need it), a **definition** or **theorem** (exactly what it says), **in simple words** (the same thing without symbols), **pictures**, and **examples** where every step and every row operation is written. Red boxes show the **mistakes students really make**, with the correct way next to them.



Colour code in every picture. Blue = the original vector, orange = the vector after multiplying by A, gold = an eigenvector (the arrow that does not turn), teal = an eigenspace.

Book references. Anton = *Elementary Linear Algebra* (Anton & Rorres), Sections 5.1 and 5.2. Williams = *Linear Algebra with Applications* (Gareth Williams), Section 3.4. Example numbers are the same as in the books. Every number was checked by the Python file `eigen_lecture.py`. Solved book exercises are in the separate file *Solved Exercises*.

Two parts, two videos. Part A (Eigenvalues and Eigenvectors) goes with Video 1; Part B (Diagonalization) goes with Video 2.

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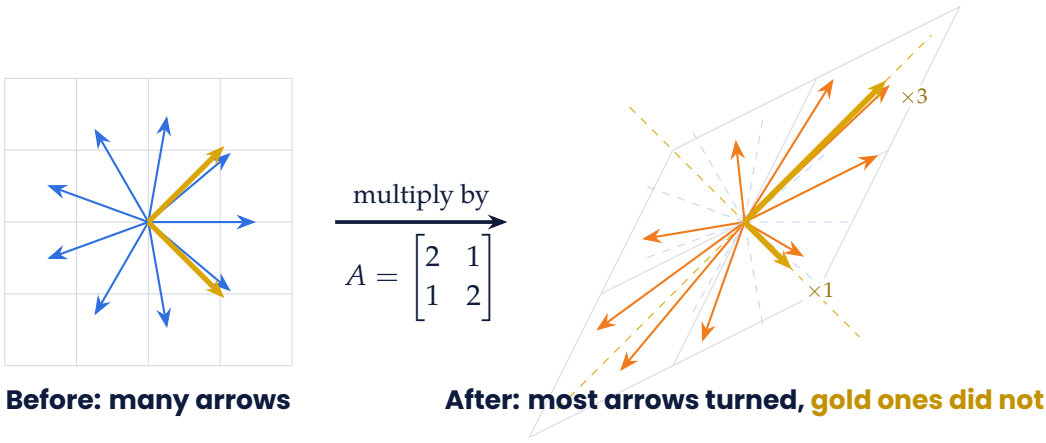
Part A Eigenvalues and Eigenvectors

1 The story: the arrows that refuse to turn

Story: a rubber sheet and one stubborn arrow

Draw many arrows from one point on a rubber sheet. Now hold the sheet at two corners and pull. Almost every arrow **turns**: it ends up pointing in a new direction and has a new length. But if you look carefully, a few arrows do something very special. They do **not** turn at all. They stay on the same line through the centre. They only become **longer**, or **shorter**, or they flip to point the opposite way.

Those stubborn arrows are called **eigenvectors** (German *eigen* = "own": the matrix's *own* directions). The number that tells us *how much* such an arrow was stretched is called its **eigenvalue**.



Before: many arrows **After: most arrows turned, gold ones did not**

Why do we care? Because these special directions make a hard matrix behave like a simple number. Along an eigenvector, "multiply by the matrix A" is the same as "multiply by the number λ". So questions about A (what happens after 100 steps? is the system stable? which page is most important?) become questions about a few numbers.

Where is this used?

- **Google search** ranks pages using the eigenvector of a huge matrix (eigenvalue 1).
- **Bridges and buildings**: engineers compute eigenvalues to find the natural vibration frequencies, so the wind cannot shake a bridge to pieces.
- **Face recognition** uses "eigenfaces": eigenvectors of a matrix built from many photos.
- **Population models** (Leslie matrices) and **weather models** (Markov chains): the long-run answer is an eigenvector.

2 Eigenvalue and eigenvector: the definition

Definition Anton §5.1, Def. 1 | Williams §3.4

Let A be an $n \times n$ matrix. A **nonzero** vector $\mathbf{x}$ in $\mathbb{R}^n$ is called an **eigenvector** of A if $A\mathbf{x}$ is a scalar multiple of $\mathbf{x}$, that is

$$\boxed{A\mathbf{x} = \lambda\mathbf{x}}$$

for some scalar λ. The scalar λ is called an **eigenvalue** of A, and $\mathbf{x}$ is said to be an eigenvector *corresponding to* λ.

In simple words

An eigenvector is an arrow that the matrix does *not* turn. The matrix only stretches it, shrinks it, or flips it. The eigenvalue is the stretching factor.

Read the equation slowly. On the left, $A\mathbf{x}$ is "matrix times vector" (hard work). On the right, $\lambda\mathbf{x}$ is "number times vector" (easy). An eigenvector is exactly a vector for which the hard work and the easy work give the same answer.

What the size of λ tells you



$0 < \lambda < 1$ shrinks | $\lambda > 1$ stretches | $-1 < \lambda < 0$ shrinks and flips | $\lambda < -1$ stretches and flips

In every case the orange arrow $\lambda\mathbf{x}$ stays on the *same line* as the blue arrow $\mathbf{x}$. That is the whole meaning of "eigen".

Example 1 Anton §5.1, Example 1**: checking an eigenvector**

Show that $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$.

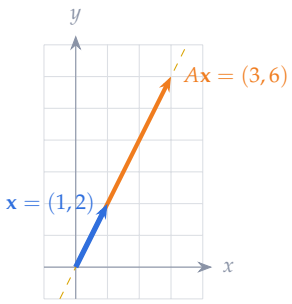
Multiply row by column

$$A\mathbf{x} = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} (3)(1) + (0)(2) \\ (8)(1) + (-1)(2) \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Compare with $\mathbf{x}$

$\begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3\mathbf{x}$. So $A\mathbf{x} = 3\mathbf{x}$: $\mathbf{x}$ is an eigenvector and its eigenvalue is $\lambda = 3$.

Geometrically, A took the arrow from $(1, 2)$ to $(3, 6)$: same line, three times as long.



Common mistake: the zero vector

"$\mathbf{0}$ is an eigenvector because $A\mathbf{0} = \lambda\mathbf{0}$." **Wrong.** $A\mathbf{0} = \lambda\mathbf{0}$ is true for *every* A and *every* λ, so it tells us nothing. That is why the definition says an eigenvector must be **nonzero**. But an **eigenvalue may be zero**: $\lambda = 0$ is allowed (see Section 7).

Common mistake: ``an eigenvector'' is never just one vector

If $\mathbf{x}$ is an eigenvector for λ, then so is $2\mathbf{x}$, $-5\mathbf{x}$, $\frac{1}{3}\mathbf{x}$, ... (any nonzero multiple), because $A(c\mathbf{x}) = cA\mathbf{x} = c\lambda\mathbf{x} = \lambda(c\mathbf{x})$. So the book's answer $\begin{bmatrix}1\\1\end{bmatrix}$ and your answer $\begin{bmatrix}3\\3\end{bmatrix}$ are **both correct**. Do not panic if your eigenvector looks different from the book.

3 How to find eigenvalues: the characteristic equation

3.1 Where the equation comes from

We cannot guess eigenvectors, so we turn the definition into an equation we can solve. Follow the four lines:

$$A\mathbf{x} = \lambda\mathbf{x} \implies A\mathbf{x} = \lambda I\mathbf{x} \implies \lambda I\mathbf{x} - A\mathbf{x} = \mathbf{0} \implies (\lambda I - A)\,\mathbf{x} = \mathbf{0}$$

(We wrote $\lambda\mathbf{x} = \lambda I\mathbf{x}$ because we cannot subtract a matrix from a number; λI is a matrix.)

This is a **homogeneous system**. It always has the trivial solution $\mathbf{x} = \mathbf{0}$, but we need a *nonzero* solution. A square homogeneous system has a nonzero solution **exactly when its coefficient matrix is not invertible**, i.e. when its determinant is zero.

Theorem 5.1.1 Anton

If A is an $n \times n$ matrix, then λ is an eigenvalue of A if and only if

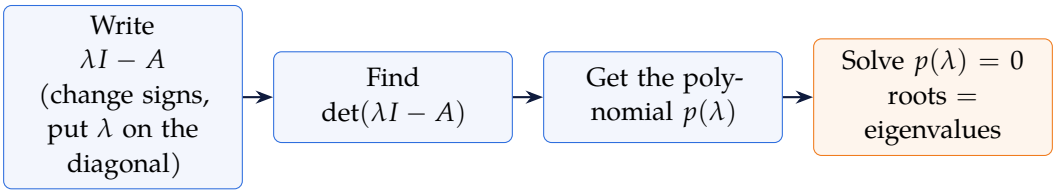
$$\det(\lambda I - A) = 0.$$

This is called the **characteristic equation** of A.

When we expand $\det(\lambda I - A)$ we get a polynomial in λ of degree n whose leading coefficient is 1:

$$p(\lambda) = \det(\lambda I - A) = \lambda^n + c_1\lambda^{n-1} + \cdots + c_n.$$

This polynomial is called the **characteristic polynomial** of A. The eigenvalues are its roots.



In simple words

The characteristic polynomial is a "detector". You feed it a number λ. If the answer is 0, that number is an eigenvalue. If not, it is not.

Example 2 Anton §5.1, Example 2 **: all eigenvalues of a 2×2 matrix**

Find all eigenvalues of $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$.

Step 1: form $\lambda I - A$

$$\lambda I - A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} = \begin{bmatrix} \lambda - 3 & 0 \\ -8 & \lambda + 1 \end{bmatrix}.$$

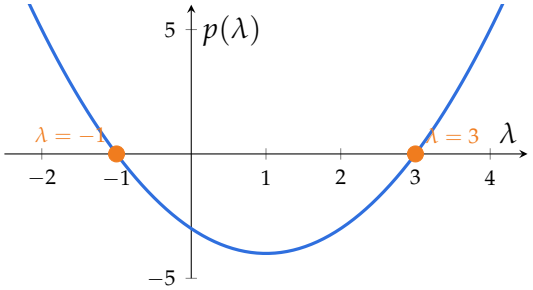
Step 2: the determinant

$$\det(\lambda I - A) = (\lambda - 3)(\lambda + 1) - (0)(-8) = (\lambda - 3)(\lambda + 1).$$

Step 3: the polynomial and its roots

$p(\lambda) = (\lambda - 3)(\lambda + 1) = \lambda^2 - 2\lambda - 3$. Setting $p(\lambda) = 0$ gives $\lambda = 3$ and $\lambda = -1$.
So besides $\lambda = 3$ (found in Example 1) the matrix has a second eigenvalue, $\lambda = -1$.

$p(\lambda) = \lambda^2 - 2\lambda - 3$ crosses zero at the eigenvalues



3.2 A quick formula for every 2×2 matrix

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - a & -b \\ -c & \lambda - d \end{vmatrix} = (\lambda - a)(\lambda - d) - bc = \lambda^2 - (\operatorname{tr} A)\,\lambda + \det A$$

because $a + d = \operatorname{tr} A$ and $ad - bc = \det A$. Check with Example 2: $\operatorname{tr} A = 3 + (-1) = 2$, $\det A = (3)(-1) - (0)(8) = -3$, so $p(\lambda) = \lambda^2 - 2\lambda - 3$. ✓

Two facts that let you check every answer. For any $n \times n$ matrix (counting repeated eigenvalues):

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \operatorname{tr} A, \qquad \lambda_1 \lambda_2 \cdots \lambda_n = \det A.$$

Example 2: $3 + (-1) = 2 = \operatorname{tr} A$ and $(3)(-1) = -3 = \det A$. ✓

Example 3 Anton §5.1, Example 3 **: a 3×3 matrix**

Find the eigenvalues of $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$.

Step 1: $\lambda I - A$

$$\lambda I - A = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -4 & 17 & \lambda - 8 \end{bmatrix}.$$

Step 2: determinant by cofactor expansion along the first row

$$\begin{aligned} \det(\lambda I - A) &= \lambda \begin{vmatrix} \lambda & -1 \\ 17 & \lambda - 8 \end{vmatrix} - (-1) \begin{vmatrix} 0 & -1 \\ -4 & \lambda - 8 \end{vmatrix} + 0 \\ &= \lambda\big[\lambda(\lambda - 8) - (-1)(17)\big] + \big[(0)(\lambda - 8) - (-1)(-4)\big] \\ &= \lambda(\lambda^2 - 8\lambda + 17) + (-4) = \lambda^3 - 8\lambda^2 + 17\lambda - 4. \end{aligned}$$

Step 3: look for an integer root

Integer roots must divide the constant term -4: try $\pm 1, \pm 2, \pm 4$. $p(1) = 1 - 8 + 17 - 4 = 6$, $p(2) = 8 - 32 + 34 - 4 = 6$, $p(4) = 64 - 128 + 68 - 4 = 0$. So $\lambda = 4$ is a root and $(\lambda - 4)$ is a factor.

Step 4: divide (synthetic division by 4)

$$\begin{array}{c|cccc} 4 & 1 & -8 & 17 & -4 \\ & & 4 & -16 & 4 \\ \hline & 1 & -4 & 1 & 0 \end{array} \qquad \Longrightarrow \qquad p(\lambda) = (\lambda - 4)(\lambda^2 - 4\lambda + 1).$$

Step 5: solve the quadratic

$$\lambda = \frac{4 \pm \sqrt{16 - 4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}.$$

Eigenvalues: $\lambda = 4, \quad 2 + \sqrt{3}, \quad 2 - \sqrt{3}$. Check: $4 + (2 + \sqrt{3}) + (2 - \sqrt{3}) = 8 = \operatorname{tr} A$ ✓.

Example Williams §3.4, Example 1**: the other sign convention**

Find the eigenvalues of $A = \begin{bmatrix} -4 & -6 \\ 3 & 5 \end{bmatrix}$.

Williams uses $\det(A - \lambda I) = 0$ ("subtract λ from the diagonal"):

$$|A - \lambda I| = \begin{vmatrix} -4 - \lambda & -6 \\ 3 & 5 - \lambda \end{vmatrix} = (-4 - \lambda)(5 - \lambda) - (-6)(3) = \lambda^2 - \lambda - 20 + 18 = \lambda^2 - \lambda - 2.$$

$\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1) = 0$, so $\lambda = 2$ and $\lambda = -1$. Check: $2 + (-1) = 1 = \operatorname{tr} A$, $(2)(-1) = -2 = \det A$ ✓.

Common mistake: "my book says $A - \lambda I$, the other says $\lambda I - A$"

Both are correct. $\det(A - \lambda I) = (-1)^n \det(\lambda I - A)$, so the two polynomials differ at most by a sign, and **the roots (the eigenvalues) are exactly the same**. Just use one convention all the way through a problem. With $\lambda I - A$ the leading coefficient is always $+1$.

Common mistake: "$\lambda I - A$" means subtract λ from every entry

Wrong. λI has λ only on the main diagonal. So $\lambda I - A$ has $\lambda - a_{ii}$ on the diagonal and $-a_{ij}$ everywhere else. Off-diagonal entries only change sign; they never get a λ.

Common mistake: real matrix, complex eigenvalues

Not every real matrix has real eigenvalues. The rotation $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (turn by 90°) has $p(\lambda) = \lambda^2 + 1$, roots $\pm i$. This makes sense: a 90° turn turns *every* arrow, so there is no real direction that stays on its line. An $n \times n$ matrix has **at most** n distinct eigenvalues.

4 Eigenvalues of triangular matrices: read them off the diagonal

Example 4 Anton §5.1, Example 4: **upper triangular**

For an upper triangular A, the matrix $\lambda I - A$ is also upper triangular, and the determinant of a triangular matrix is the product of its diagonal entries:

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & -a_{13} & -a_{14} \\ 0 & \lambda - a_{22} & -a_{23} & -a_{24} \\ 0 & 0 & \lambda - a_{33} & -a_{34} \\ 0 & 0 & 0 & \lambda - a_{44} \end{bmatrix} = (\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33})(\lambda - a_{44}).$$

So the eigenvalues are $\lambda = a_{11}, a_{22}, a_{33}, a_{44}$: **exactly the diagonal entries**.

Theorem 5.1.2 Anton

If A is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), then the eigenvalues of A are the entries on the main diagonal of A.



gold diagonal entries = the eigenvalues

Example 5 Anton §5.1, Example 5: **lower triangular, by inspection**

$$A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -1 & \frac{2}{3} & 0 \\ 5 & -8 & -\frac{1}{4} \end{bmatrix} \quad \Longrightarrow \quad \lambda = \tfrac{1}{2}, \quad \lambda = \tfrac{2}{3}, \quad \lambda = -\tfrac{1}{4}.$$

No determinant is needed: the matrix is lower triangular, so its eigenvalues are its diagonal entries.

Common mistake: using the diagonal of a matrix that is *not* triangular

For $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$ (lower triangular) the diagonal does give the eigenvalues $3, -1$. But for $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ the eigenvalues are 2 and -3 (Example 6), **not** -1 and 0. The shortcut works *only* for triangular matrices.

Common mistake: row reducing first

"I will row reduce A to a triangular matrix and then read the eigenvalues." **Wrong!** Row operations change the eigenvalues. Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has eigenvalues $3, -1$; after $R_2 \to R_2 - 2R_1$ we get $\begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}$ with "eigenvalues" $1, -3$. Row reduction is used later on $\lambda I - A$ (to find eigenvectors), never on A to find eigenvalues.

5 Four ways to say "λ is an eigenvalue"

Theorem 5.1.3 Anton

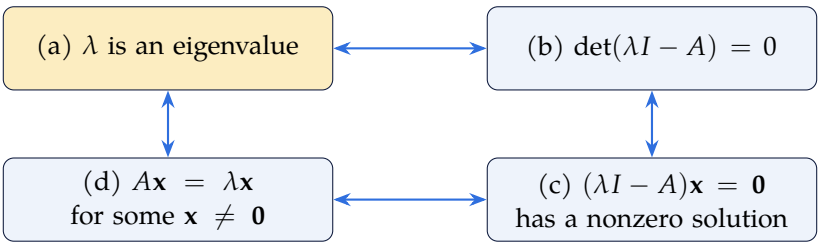
If A is an $n \times n$ matrix, the following statements are equivalent.

(a) λ is an eigenvalue of A.

(b) λ is a solution of the characteristic equation $\det(\lambda I - A) = 0$.

(c) The system of equations $(\lambda I - A)\mathbf{x} = \mathbf{0}$ has nontrivial solutions.

(d) There is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$.



In simple words

Whichever of the four you can check most easily, check that one. They always agree.

6 Finding eigenvectors: eigenspaces and their bases

Once we know an eigenvalue λ, its eigenvectors are the **nonzero** solutions of

$$(\lambda I - A)\mathbf{x} = \mathbf{0}.$$

Definition Anton §5.1 | Williams §3.4, Thm 3.9

The solution space of $(\lambda I - A)\mathbf{x} = \mathbf{0}$ is called the **eigenspace** of A corresponding to λ. It is

1. the null space of the matrix $\lambda I - A$;
2. the set of all eigenvectors for λ, *together with* the zero vector.

It is a **subspace** of $\mathbb{R}^n$ (Williams, Theorem 3.9): if $A\mathbf{x}_1 = \lambda\mathbf{x}_1$ and $A\mathbf{x}_2 = \lambda\mathbf{x}_2$ then $A(\mathbf{x}_1 + \mathbf{x}_2) = \lambda(\mathbf{x}_1 + \mathbf{x}_2)$ and $A(c\mathbf{x}_1) = \lambda(c\mathbf{x}_1)$.

In simple words

An eigenspace is the whole "family" of directions that one eigenvalue keeps in place. In $\mathbb{R}^2$ it is a line through the origin; in $\mathbb{R}^3$ it can be a line or a whole plane.

Take one eigenvalue λ → Write $\lambda I - A$ (numbers only) → Row reduce $[\lambda I - A \mid \mathbf{0}]$ → Free variables → parameters s, t → Split into vectors: those are the basis

Example 6 Anton §5.1, Example 6 **: bases for eigenspaces,** 2×2

Find bases for the eigenspaces of $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$.

Eigenvalues

$\det(\lambda I - A) = \begin{vmatrix} \lambda+1 & -3 \\ -2 & \lambda \end{vmatrix} = (\lambda+1)\lambda - (-3)(-2) = \lambda^2 + \lambda - 6 = (\lambda - 2)(\lambda + 3)$. So $\lambda = 2$ and $\lambda = -3$.

Eigenspace for $\lambda = 2$

$2I - A = \begin{bmatrix} 2+1 & -3 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ -2 & 2 \end{bmatrix}$. Row reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 3 & -3 & 0 \\ -2 & 2 & 0 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ -2 & 2 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + 2R_1} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

So $x_1 - x_2 = 0$; x_2 is free. Put $x_2 = t$: $x_1 = t$. $\mathbf{x} = \begin{bmatrix} t \\ t \end{bmatrix} = t\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Basis: $\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$.

Eigenspace for $\lambda = -3$

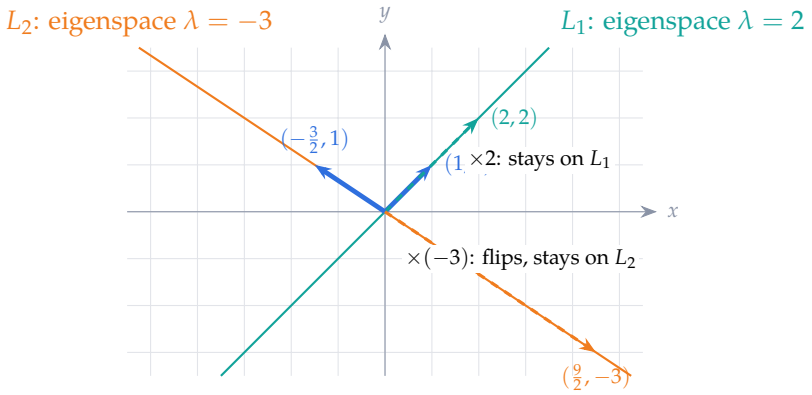
$-3I - A = \begin{bmatrix} -3+1 & -3 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} -2 & -3 \\ -2 & -3 \end{bmatrix}$.

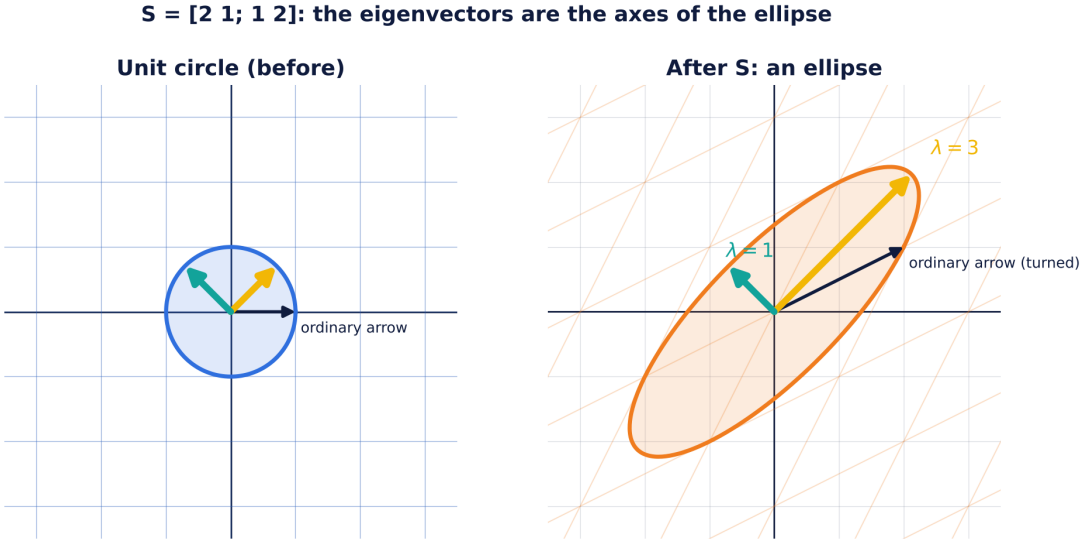
$$\left[\begin{array}{cc|c} -2 & -3 & 0 \\ -2 & -3 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{cc|c} -2 & -3 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{1}{2}R_1} \left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 0 & 0 \end{array}\right].$$

So $x_1 = -\frac{3}{2}x_2$. Put $x_2 = t$: $\mathbf{x} = t\begin{bmatrix} -\frac{3}{2} \\ 1 \end{bmatrix}$. Basis: $\left\{\begin{bmatrix} -3/2 \\ 1 \end{bmatrix}\right\}$ (or $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$, a nicer multiple).

Check

$A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1+3 \\ 2+0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ✓ $A\begin{bmatrix} -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3+6 \\ -6+0 \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \end{bmatrix} = -3\begin{bmatrix} -3 \\ 2 \end{bmatrix}$ ✓





From the Python file: the unit circle becomes an ellipse; only along the eigen-directions does an arrow keep its line.

Example 7 Anton §5.1, Example 7: a 3×3 matrix, one eigenspace is a plane

Find bases for the eigenspaces of $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$.

Eigenvalues

Expand $\det(\lambda I - A)$ along the second column (it has two zeros):

$$\det \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda - 2 & -1 \\ -1 & 0 & \lambda - 3 \end{bmatrix} = (\lambda - 2) \begin{vmatrix} \lambda & 2 \\ -1 & \lambda - 3 \end{vmatrix} = (\lambda - 2)\big[\lambda(\lambda - 3) - (2)(-1)\big]$$

$$= (\lambda - 2)(\lambda^2 - 3\lambda + 2) = (\lambda - 2)(\lambda - 1)(\lambda - 2) = (\lambda - 1)(\lambda - 2)^2.$$

(Expanded: $\lambda^3 - 5\lambda^2 + 8\lambda - 4$.) Distinct eigenvalues: $\lambda = 1$ and $\lambda = 2$ (twice).

Eigenspace for $\lambda = 2$

$$2I - A = \begin{bmatrix} 2 & 0 & 2 \\ -1 & 0 & -1 \\ -1 & 0 & -1 \end{bmatrix}.$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 2 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

One equation $x_1 + x_3 = 0$; x_2 and x_3 are free. Put $x_2 = t$, $x_3 = s$: $x_1 = -s$.

$$\mathbf{x} = \begin{bmatrix} -s \\ t \\ s \end{bmatrix} = s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \qquad \text{Basis: } \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \text{ (a plane).}$$

Eigenspace for $\lambda = 1$

$$1I - A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & -1 & -1 \\ -1 & 0 & -2 \end{bmatrix}.$$

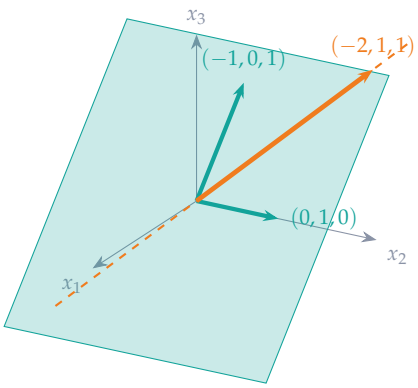
$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ -1 & -1 & -1 & 0 \\ -1 & 0 & -2 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_2 \to -R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

$x_1 + 2x_3 = 0$, $x_2 - x_3 = 0$; x_3 free. Put $x_3 = s$: $x_1 = -2s$, $x_2 = s$.

$$\mathbf{x} = s \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}. \qquad \text{Basis: } \left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\} \text{ (a line).}$$

Check one of them

$$A \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0+0-2 \\ -2+2+1 \\ -2+0+3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \checkmark$$



Teal plane: eigenspace for $\lambda = 2$ (dimension 2). Every vector in this plane is just doubled.

Orange line: eigenspace for $\lambda = 1$ (dimension 1). Every vector on it is left exactly as it is.

Example Williams §3.4, Example 2**: repeated eigenvalue, plane again**

For $A = \begin{bmatrix} 5 & 4 & 2 \\ 4 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$ Williams finds $|A - \lambda I| = -(\lambda - 10)(\lambda - 1)^2$, so $\lambda = 10$ and $\lambda = 1$ (repeated). For $\lambda = 10$: $A - 10I = \begin{bmatrix} -5 & 4 & 2 \\ 4 & -5 & 2 \\ 2 & 2 & -8 \end{bmatrix}$ gives $x_1 = 2r$, $x_2 = 2r$, $x_3 = r$, basis $\left\{ \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right\}$ (a line). For $\lambda = 1$: $A - I = \begin{bmatrix} 4 & 4 & 2 \\ 4 & 4 & 2 \\ 2 & 2 & 1 \end{bmatrix}$; after $R_2 \to R_2 - R_1$, $R_3 \to R_3 - \frac{1}{2}R_1$ only $4x_1 + 4x_2 + 2x_3 = 0$ remains, i.e. $x_3 = -2x_1 - 2x_2$. With $x_1 = s$, $x_2 = t$: basis $\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$ (a plane). Check: $A \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 10+8+2 \\ 8+10+2 \\ 4+4+2 \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \\ 10 \end{bmatrix} = 10 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ ✓.

Common mistake: row reducing A instead of $\lambda I - A$

The eigenvectors are the null space of $\lambda I - A$ (a *different* matrix for each λ), not the null space of A. Always substitute the number λ first, then row reduce.

Common mistake: "I got only the zero solution"

If $(\lambda I - A)\mathbf{x} = \mathbf{0}$ gives only $\mathbf{x} = \mathbf{0}$, then your λ is **not** an eigenvalue: there is an arithmetic mistake in the characteristic polynomial. For a true eigenvalue the reduced matrix *always* has at least one row of zeros.

Common mistake: writing the eigenspace as one vector

The eigenspace is a whole subspace. A *basis* for it is a short list of vectors. Say "basis: $\{(1,1)\}$" or "eigenspace: all $t(1,1)$". And remember: dimension of an eigenspace = number of free variables.

7 Eigenvalues and invertibility

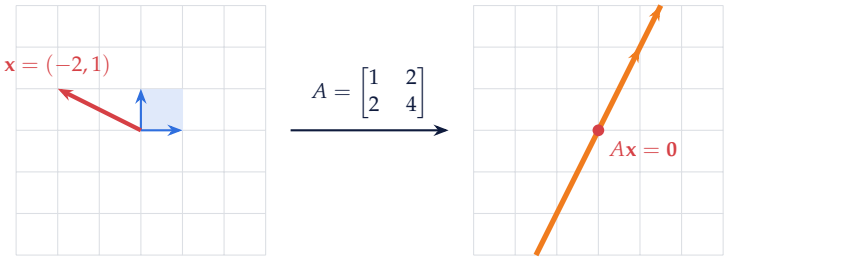
Theorem 5.1.4 Anton

A square matrix A is invertible if and only if $\lambda = 0$ is **not** an eigenvalue of A.

Why? Put $\lambda = 0$ in the characteristic polynomial: $p(0) = \det(0 \cdot I - A) = \det(-A) = (-1)^n \det A$. So

$$0 \text{ is an eigenvalue} \iff p(0) = 0 \iff \det A = 0 \iff A \text{ is not invertible.}$$

Another way to see it: $\lambda = 0$ is an eigenvalue means $A\mathbf{x} = 0 \cdot \mathbf{x} = \mathbf{0}$ for some $\mathbf{x} \neq \mathbf{0}$, so A sends a whole line to the single point $\mathbf{0}$. Nothing can undo that.



before after: the whole plane is squashed onto a line

Here $\det A = (1)(4) - (2)(2) = 0$, $p(\lambda) = \lambda^2 - 5\lambda$, eigenvalues 0 and 5. The vector $(-2, 1)$ is an eigenvector for $\lambda = 0$: $A\mathbf{x} = \begin{bmatrix} -2+2 \\ -4+4 \end{bmatrix} = \mathbf{0}$.

Example 8 Anton §5.1, Example 8

The matrix of Example 7 has eigenvalues 1 and 2; neither is 0, so it is invertible. Check: expand $\det A$ along the second column: $\det A = 2\begin{vmatrix} 0 & -2 \\ 1 & 3 \end{vmatrix} = 2\big[(0)(3) - (-2)(1)\big] = 2(2) = 4 \neq 0$ ✓. Also $1 \cdot 2 \cdot 2 = 4 = \det A$.

Common mistake: "$\lambda = 0$ is not allowed"

Only the zero *vector* is forbidden. The *number* 0 can be an eigenvalue, and when it is, the matrix is singular. Bonus fact: if A is invertible and $A\mathbf{x} = \lambda\mathbf{x}$, then $A^{-1}\mathbf{x} = \frac{1}{\lambda}\mathbf{x}$ (same eigenvector, eigenvalue flipped to $1/\lambda$).

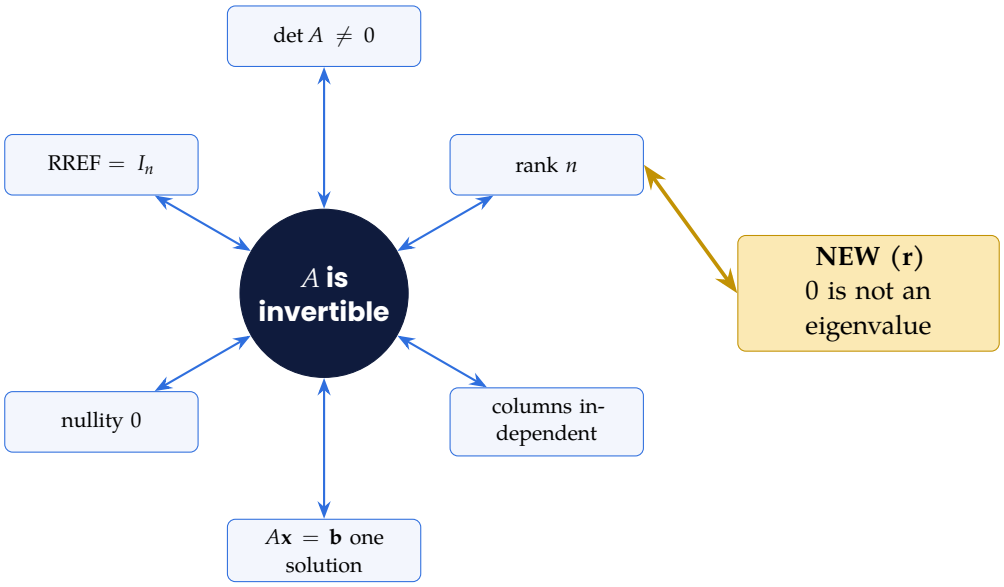
8 More on the Equivalence Theorem

Theorem 5.1.4 adds one more line to the big list of equivalent statements we have been building since Chapter 1.

Theorem 5.1.5 Anton: Equivalent statements

If A is an $n \times n$ matrix, the following are equivalent:

(a) A is invertible.
(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
(c) The reduced row echelon form of A is I_n.
(d) A is a product of elementary matrices.
(e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$.
(f) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$.
(g) $\det(A) \neq 0$.
(h) The columns of A are linearly independent.
(i) The rows of A are linearly independent.
(j) The columns of A span $\mathbb{R}^n$.
(k) The rows of A span $\mathbb{R}^n$.
(l) The columns of A form a basis for $\mathbb{R}^n$.
(m) The rows of A form a basis for $\mathbb{R}^n$.
(n) A has rank n.
(o) A has nullity 0.
(p) The orthogonal complement of the null space of A is $\mathbb{R}^n$.
(q) The orthogonal complement of the row space of A is $\{\mathbf{0}\}$.
(r) $\lambda = 0$ is not an eigenvalue of A.



In simple words

All eighteen sentences are "the same sentence in different clothes". If one is true, all are true; if one is false, all are false. So you can test invertibility with whichever one is easiest, now including "is 0 an eigenvalue?"

Part B Diagonalization

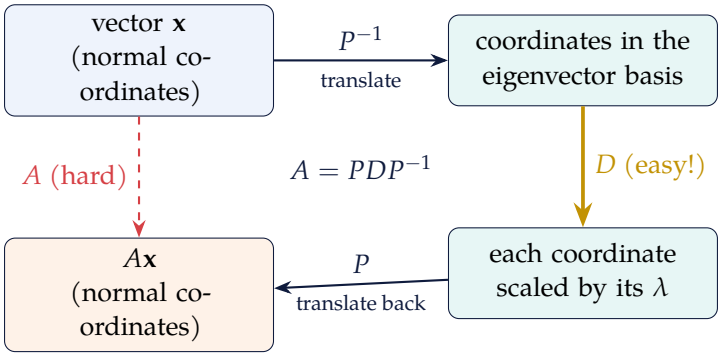
9 Why diagonalize? Similar matrices

Story: the easiest matrices in the world

Which matrix is easiest to work with? A **diagonal** one. Multiply it by a vector: each coordinate is just scaled. Raise it to the 100th power: just raise each diagonal number to the 100th power.

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \quad \Longrightarrow \quad D^{100} = \begin{bmatrix} 2^{100} & 0 \\ 0 & 3^{100} \end{bmatrix}.$$

Most matrices are not diagonal. But here is the magic: **if we look at a matrix through the eyes of its eigenvectors, it becomes diagonal.** Choosing those eyes is called *diagonalization*. It is like translating a hard sentence into a language where it becomes easy, doing the work there, and translating back.



Definition 1 Anton §5.2: **similar matrices**

If A and B are square matrices, then B is **similar** to A if there is an invertible matrix P such that

$$B = P^{-1}AP.$$

(Then also $A = PBP^{-1} = Q^{-1}BQ$ with $Q = P^{-1}$, so A is similar to B too; we simply say A and B are *similar*.)

In simple words

Similar matrices are the *same transformation* described in two different coordinate systems. That is why they share everything that does not depend on the coordinates.

What similar matrices have in common (similarity invariants, Anton Table 1):

Property	A and $P^{-1}AP$ ...
Determinant	have the same determinant
Invertibility	are both invertible or both not
Rank and nullity	have the same rank and the same nullity
Trace	have the same trace
Characteristic polynomial	have the same characteristic polynomial
Eigenvalues	have the same eigenvalues
Eigenspace dimension	have eigenspaces of the same dimension for each λ

For example $\det(P^{-1}AP) = \det(P^{-1})\det(A)\det(P) = \dfrac{1}{\det P}\det(A)\det(P) = \det(A)$.

Common mistake: similar does *not* mean equal eigenvectors

Similar matrices have the same eigenvalues, but generally **different eigenvectors** (if $A\mathbf{x} = \lambda\mathbf{x}$ then $B = P^{-1}AP$ has eigenvector $P^{-1}\mathbf{x}$). Also, same eigenvalues does not guarantee similar: $I = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ and $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ both have only $\lambda = 1$, yet they are not similar (the only matrix similar to I is I itself).

Definition 2 Anton §5.2 **: diagonalizable**

A square matrix A is **diagonalizable** if it is similar to some diagonal matrix, that is, if there is an invertible matrix P such that $P^{-1}AP$ is diagonal. In this case P is said to **diagonalize** A.

10 When is a matrix diagonalizable?

Theorem 5.2.1 Anton

If A is an $n \times n$ matrix, the following statements are equivalent.

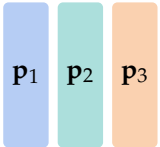
(a) A is diagonalizable.

(b) A has n linearly independent eigenvectors.

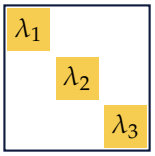
The idea of the proof in one picture. Put the eigenvectors $\mathbf{p}_1, \ldots, \mathbf{p}_n$ as the columns of P and the eigenvalues on the diagonal of D. Then, column by column,

$$AP = A\begin{bmatrix}\mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n\end{bmatrix} = \begin{bmatrix}A\mathbf{p}_1 & A\mathbf{p}_2 & \cdots & A\mathbf{p}_n\end{bmatrix} = \begin{bmatrix}\lambda_1\mathbf{p}_1 & \lambda_2\mathbf{p}_2 & \cdots & \lambda_n\mathbf{p}_n\end{bmatrix} = PD.$$

If the columns are linearly independent, P is invertible and $AP = PD$ becomes $P^{-1}AP = D$.



P: **eigenvectors (columns)**



D: **eigenvalues**

Column i of P and diagonal entry i of D **must match**: $\mathbf{p}_i$ is an eigenvector for λ_i.

Theorem 5.2.2 Anton

(a) If $\lambda_1, \ldots, \lambda_k$ are **distinct** eigenvalues of A and $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are corresponding eigenvectors, then $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent.

(b) An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

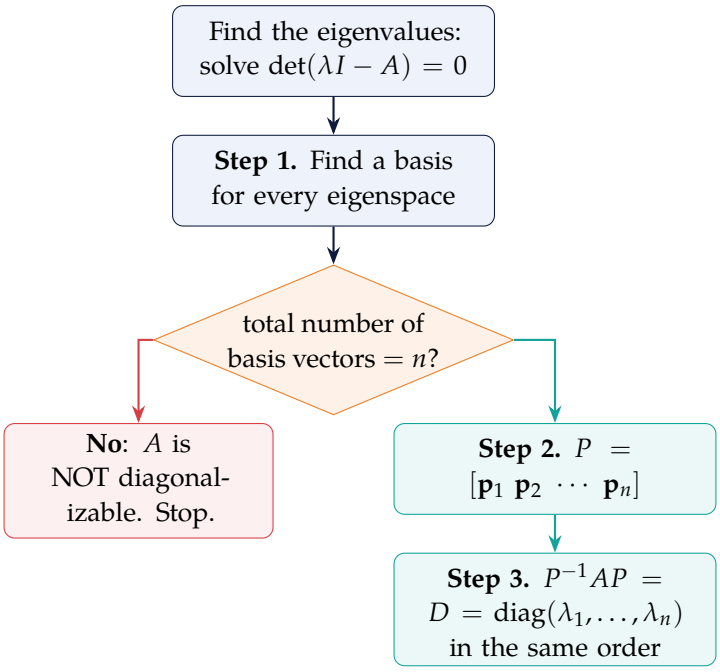
In simple words

Eigenvectors that belong to *different* eigenvalues can never lie on the same line (or in the span of each other). So if you find n different eigenvalues, you automatically get n independent eigenvectors, and the matrix is diagonalizable. No checking needed!

Common mistake: reading Theorem 5.2.2(b) backwards

"n distinct eigenvalues $\Rightarrow$ diagonalizable" is true. "Diagonalizable $\Rightarrow$ n distinct eigenvalues" is **false**. The identity matrix I_3 is already diagonal (so diagonalizable) but has only one eigenvalue. Repeated eigenvalues do *not* automatically mean "not diagonalizable": you must check the eigenspaces (Section 17).

11 The procedure for diagonalizing a matrix



Example 1 Anton §5.2, Example 1: finding P that diagonalizes A

Find a matrix P that diagonalizes $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$.

Eigenvalues and eigenspaces (done in Part A, Example 7)

$p(\lambda) = (\lambda - 1)(\lambda - 2)^2$, and

$$\lambda = 2: \ \mathbf{p}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \ \mathbf{p}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \qquad \lambda = 1: \ \mathbf{p}_3 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}.$$

Step 1: count

$2 + 1 = 3$ basis vectors and $n = 3$, so A **is** diagonalizable.

Step 2: build P and D (same order!)

$$P = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 3a: find P^{-1} by row reducing $[P \mid I]$

$$\left[\begin{array}{ccc|ccc} -1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to -R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \end{array}\right] \xrightarrow[R_2 \to R_2 - R_3]{R_1 \to R_1 - 2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 & 0 & -1 \end{array}\right].$$

So $P^{-1} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix}$.

Step 3b: check $P^{-1}AP = D$

First AP (each column is $A\mathbf{p}_i = \lambda_i \mathbf{p}_i$):

$$AP = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & -2 \\ 0 & 2 & 1 \\ 2 & 0 & 1 \end{bmatrix} \quad (= [2\mathbf{p}_1 \ \ 2\mathbf{p}_2 \ \ 1\mathbf{p}_3]).$$

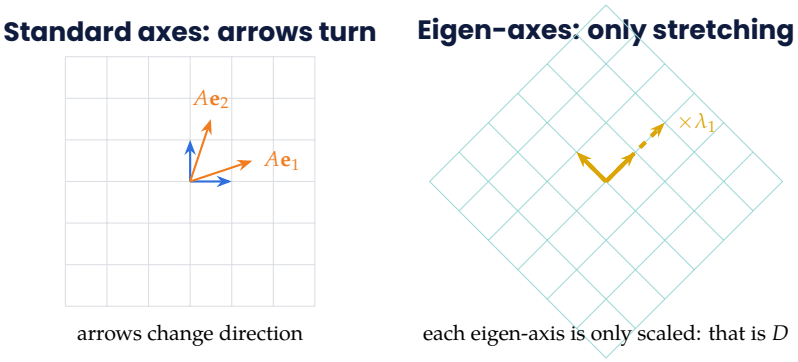
Then

$$P^{-1}(AP) = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} -2 & 0 & -2 \\ 0 & 2 & 1 \\ 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2+0+4 & 0+0+0 & -2+0+2 \\ -2+0+2 & 0+2+0 & -2+1+1 \\ 2+0-2 & 0+0+0 & 2+0-1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = D \ \checkmark$$

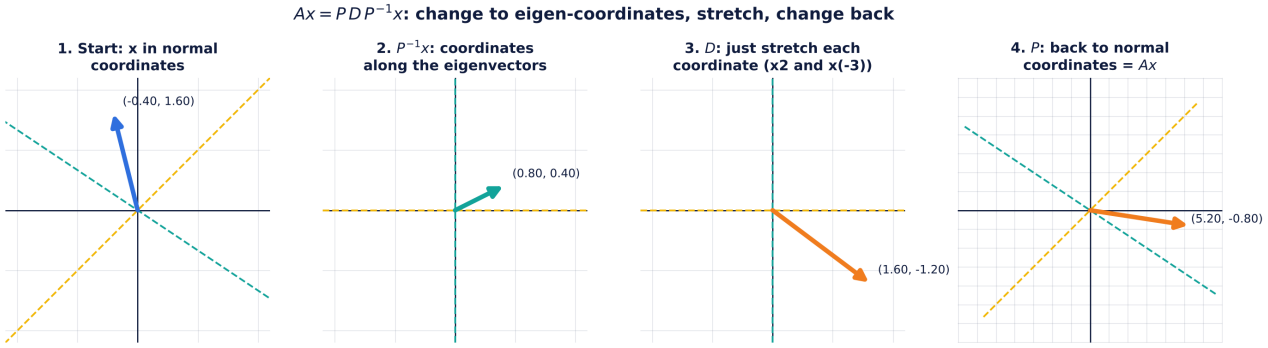
Common mistake: mixing the order

If you change the order of the columns of P, you must change the order of the eigenvalues in D in the same way. With $P = [\mathbf{p}_1 \ \mathbf{p}_3 \ \mathbf{p}_2]$ you get $D = \operatorname{diag}(2, 1, 2)$. Both are correct; mismatched orders are wrong.



arrows change direction — each eigen-axis is only scaled: that is D

(Illustrated with $A = \begin{bmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{bmatrix}$: eigenvectors $(1,1)$ with $\lambda = 2$ and $(-1,1)$ with $\lambda = 1$.)



From the Python file: $Ax = P\,D\,P^{-1}x$ in four steps: translate into eigen-coordinates, scale, translate back.

12 A matrix that is not diagonalizable

Example 2 Anton §5.2, Example 2

Show that $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{bmatrix}$ is not diagonalizable.

Eigenvalues

A is lower triangular, so $\det(\lambda I - A) = \begin{vmatrix} \lambda-1 & 0 & 0 \\ -1 & \lambda-2 & 0 \\ 3 & -5 & \lambda-2 \end{vmatrix} = (\lambda-1)(\lambda-2)^2$. Eigenvalues: 1 and 2 (twice).

Eigenspace for $\lambda = 1$

$$1I - A = \begin{bmatrix} 0 & 0 & 0 \\ -1 & -1 & 0 \\ 3 & -5 & -1 \end{bmatrix}.$$

$$\xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} -1 & -1 & 0 \\ 0 & 0 & 0 \\ 3 & -5 & -1 \end{bmatrix} \xrightarrow{R_1 \to -R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 3 & -5 & -1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -8 & -1 \end{bmatrix}$$

$$\xrightarrow{R_2\leftrightarrow R_3}\begin{bmatrix}1&1&0\\0&-8&-1\\0&0&0\end{bmatrix}\xrightarrow{R_2\to-\frac{1}{8}R_2}\begin{bmatrix}1&1&0\\0&1&\frac{1}{8}\\0&0&0\end{bmatrix}\xrightarrow{R_1\to R_1-R_2}\begin{bmatrix}1&0&-\frac{1}{8}\\0&1&\frac{1}{8}\\0&0&0\end{bmatrix}.$$

$x_1=\frac{1}{8}x_3$, $x_2=-\frac{1}{8}x_3$, $x_3=t$ free. Basis: $\left\{\begin{bmatrix}1/8\\-1/8\\1\end{bmatrix}\right\}$: **one** vector.

Eigenspace for $\lambda=2$

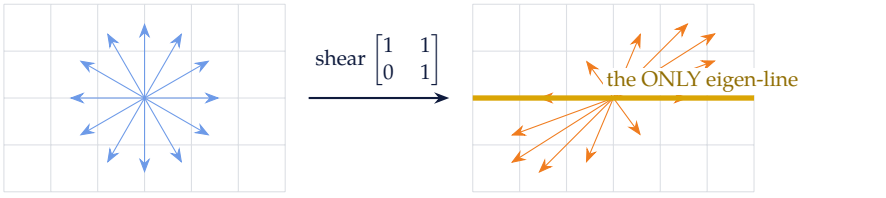
$$2I-A=\begin{bmatrix}1&0&0\\-1&0&0\\3&-5&0\end{bmatrix}\xrightarrow[R_3\to R_3-3R_1]{R_2\to R_2+R_1}\begin{bmatrix}1&0&0\\0&0&0\\0&-5&0\end{bmatrix}\xrightarrow{R_2\leftrightarrow R_3}\begin{bmatrix}1&0&0\\0&-5&0\\0&0&0\end{bmatrix}\xrightarrow{R_2\to-\frac{1}{5}R_2}\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}.$$

$x_1=0$, $x_2=0$, $x_3=t$. Basis: $\left\{\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$: again **one** vector, even though $\lambda=2$ is a double root.

Count

$1+1=2<3$. We cannot fill the three columns of P, so A is not diagonalizable.

Shortcut (no bases needed): $\dim(\text{eigenspace})=n-\operatorname{rank}(\lambda I-A)$. Here both $1I-A$ and $2I-A$ have rank 2, so each eigenspace has dimension $3-2=1$.



before **after: only the x-axis keeps its direction**

The shear $\begin{bmatrix}1&1\\0&1\end{bmatrix}$ is the simplest non-diagonalizable matrix: $\lambda=1$ is a double root, but there is only one eigen-direction, so we can never find two independent eigenvectors.

Common mistake: ``repeated eigenvalue, so not diagonalizable"

Not always! In Example 1 above, $\lambda=2$ was repeated and A *was* diagonalizable (its eigenspace was a plane). In Example 2, $\lambda=2$ was repeated and A was *not* diagonalizable (its eigenspace was only a line). The decision is made by the **dimension of the eigenspace**, never by repetition alone.

13 Recognizing diagonalizability quickly

Example 3 Anton §5.2, Example 3

In Part A (Example 3) we found that $A=\begin{bmatrix}0&1&0\\0&0&1\\4&-17&8\end{bmatrix}$ has three **distinct** eigenvalues 4, $2+\sqrt{3}$, $2-\sqrt{3}$. By Theorem 5.2.2(b), A is diagonalizable, and without computing P we already

know

$$P^{-1}AP = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2+\sqrt{3} & 0 \\ 0 & 0 & 2-\sqrt{3} \end{bmatrix}.$$

14 Diagonalizability of triangular matrices

Example 4 Anton §5.2, Example 4

The eigenvalues of a triangular matrix are its diagonal entries (Theorem 5.1.2). So a triangular matrix with **distinct** diagonal entries is diagonalizable. For example

$$A = \begin{bmatrix} -1 & 2 & 4 & 0 \\ 0 & 3 & 1 & 7 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

has eigenvalues -1, 3, 5, -2 (four different numbers for a 4×4 matrix), so it is diagonalizable.

Common mistake: ``triangular means diagonalizable"

Only when the diagonal entries are all different. The triangular matrices of Example 2 and of the shear above have a repeated diagonal entry and are **not** diagonalizable.

15 Eigenvalues of powers of a matrix

If $A\mathbf{x} = \lambda\mathbf{x}$, then

$$A^2\mathbf{x} = A(A\mathbf{x}) = A(\lambda\mathbf{x}) = \lambda(A\mathbf{x}) = \lambda(\lambda\mathbf{x}) = \lambda^2\mathbf{x}.$$

Theorem 5.2.3 Anton

If k is a positive integer, λ is an eigenvalue of A, and $\mathbf{x}$ is a corresponding eigenvector, then λ^k is an eigenvalue of A^k and $\mathbf{x}$ is a corresponding eigenvector. (Diagonalizability is *not* needed for this.)

Example 5 Anton §5.2, Example 5**: eigenvalues of A^7**

For $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{bmatrix}$ (Example 2) the eigenvalues are 1 and 2. So the eigenvalues of A^7 are $1^7 = 1$ and $2^7 = 128$, with the **same** eigenvectors $(\frac{1}{8}, -\frac{1}{8}, 1)$ and $(0, 0, 1)$.

15.1 Computing high powers with diagonalization

If $P^{-1}AP = D$, then $A = PDP^{-1}$ and

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD\underbrace{P^{-1}P}_{I}DP^{-1} = PD^2P^{-1}, \qquad \text{and in general} \qquad A^k = PD^kP^{-1},$$

where D^k just has $\lambda_1^k, \ldots, \lambda_n^k$ on its diagonal. All the inner $P^{-1}P$ pairs cancel, like a chain of handshakes.

Example 6 Anton §5.2, Example 6: A^{13}

Find A^{13} for $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$.

Use P, D, P^{-1} from Example 1

$$A^{13} = PD^{13}P^{-1} = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2^{13} & 0 & 0 \\ 0 & 2^{13} & 0 \\ 0 & 0 & 1^{13} \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix}, \qquad 2^{13} = 8192.$$

First PD^{13} (multiply column i of P by the ith diagonal entry)

$$PD^{13} = \begin{bmatrix} -8192 & 0 & -2 \\ 0 & 8192 & 1 \\ 8192 & 0 & 1 \end{bmatrix}.$$

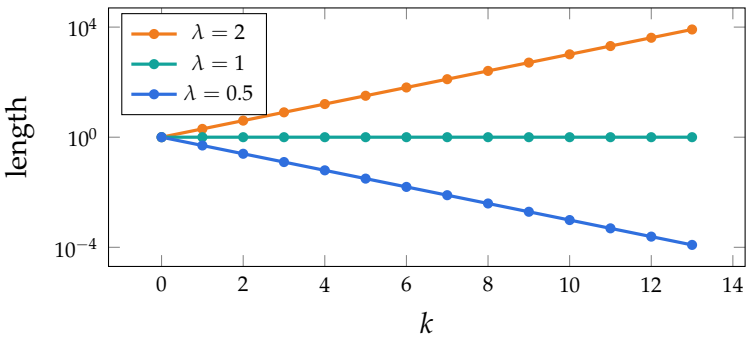
Then $(PD^{13})P^{-1}$

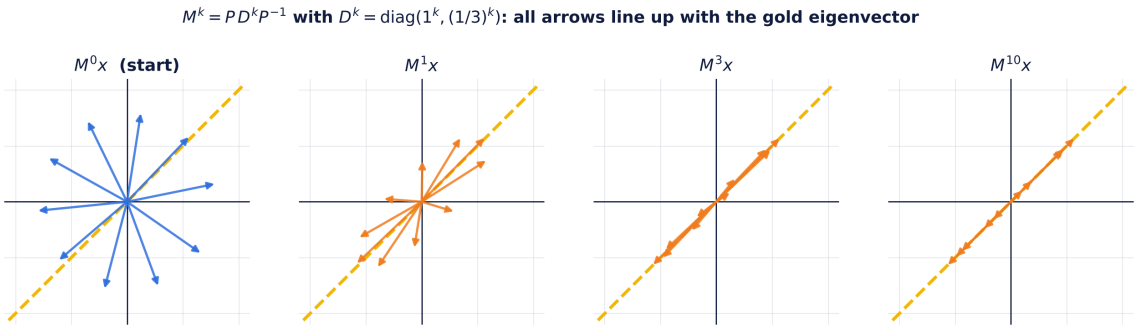
$$A^{13} = \begin{bmatrix} -8192 & 0 & -2 \\ 0 & 8192 & 1 \\ 8192 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -8192+0+2 & 0 & -16384+0+2 \\ 0+8192-1 & 8192 & 0+8192-1 \\ 8192+0-1 & 0 & 16384+0-1 \end{bmatrix}$$

$$= \begin{bmatrix} -8190 & 0 & -16382 \\ 8191 & 8192 & 8191 \\ 8191 & 0 & 16383 \end{bmatrix}.$$

Thirteen multiplications of 3×3 matrices replaced by two! For A^{1000} just change 13 to 1000.

λ^k: an eigenvalue > 1 explodes, $= 1$ stays, < 1 dies away





From the Python file: after many multiplications by A, every vector turns towards the eigenvector of the largest eigenvalue.

Common mistake: "A^k means raise every entry to the power k"

Wrong (unless A is diagonal). $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, not $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Only D^k may be computed entry by entry, which is exactly why we diagonalize.

16 Geometric and algebraic multiplicity

Definition Anton §5.2

Let λ_0 be an eigenvalue of an $n \times n$ matrix A.

- The **algebraic multiplicity** of λ_0 is the number of times $(\lambda - \lambda_0)$ appears as a factor of the characteristic polynomial.
- The **geometric multiplicity** of λ_0 is the dimension of its eigenspace (the number of free variables in $(\lambda_0 I - A)\mathbf{x} = \mathbf{0}$).

Theorem 5.2.4 Anton: geometric and algebraic multiplicity

If A is a square matrix, then:

(a) For every eigenvalue of A, the geometric multiplicity is less than or equal to the algebraic multiplicity.

(b) A is diagonalizable if and only if its characteristic polynomial can be written as a product of linear factors, and the geometric multiplicity of every eigenvalue equals its algebraic multiplicity.

In simple words

Algebraic multiplicity = how many seats an eigenvalue *booked*. Geometric multiplicity = how many independent eigenvectors actually *showed up*. A matrix is diagonalizable exactly when every booked seat is filled.

Example 1 (diagonalizable)

$\lambda = 1$ alg 1 geo 1

$\lambda = 2$ alg 2 geo 2

all seats filled ✓

Example 2 (not diagonalizable)

$\lambda = 1$ alg 1 geo 1

$\lambda = 2$ alg 2 geo 1

one seat empty ✗

Both examples have $p(\lambda) = (\lambda - 1)(\lambda - 2)^2$, so $\lambda = 2$ has algebraic multiplicity 2 in both. In Example 1 its eigenspace was a plane (geometric multiplicity 2), in Example 2 only a line (geometric multiplicity 1). That single difference decides everything.

17 The converse of Theorem 5.2.2(b) is false

Example 7 Anton §5.2, Example 7

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Both are triangular with all diagonal entries 1, so each has only **one** eigenvalue, $\lambda = 1$ (algebraic multiplicity 3).

For I

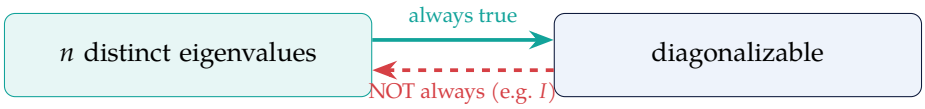
$1 \cdot I - I = 0$ (the zero matrix): every $\mathbf{x}$ solves $0\mathbf{x} = \mathbf{0}$. Eigenspace $= \mathbb{R}^3$, dimension 3. Geometric $=$ algebraic $= 3$: I **is** diagonalizable (it is already diagonal).

For J

$1 \cdot I - J = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ gives $-x_2 = 0$ and $-x_3 = 0$, $x_1 = t$ free. Eigenspace $= \left\{ t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$, dimension $1 < 3$: J is **not** diagonalizable.

Conclusion

Two 3×3 matrices with fewer than three distinct eigenvalues: one diagonalizable, one not. So "diagonalizable $\Rightarrow$ n distinct eigenvalues" is **false**.



Part C Mistakes, summary and practice

18 The ten most common student mistakes

	What students write	What is correct
1	"**0** is an eigenvector."	Eigenvectors are nonzero. The *eigenvalue* may be 0.
2	$\lambda I - A$: subtract λ from every entry.	Only the diagonal gets λ; off-diagonal entries just change sign.
3	Read eigenvalues off the diagonal of any matrix.	Only for triangular (or diagonal) matrices.
4	Row reduce A, then read eigenvalues.	Row operations change eigenvalues. Use $\det(\lambda I - A) = 0$.
5	Find eigenvectors from the null space of A.	Null space of $\lambda I - A$, a different matrix for each λ.
6	"My eigenvector differs from the book, so it is wrong."	Any nonzero multiple is correct.
7	"Repeated eigenvalue $\Rightarrow$ not diagonalizable."	Check the dimension of the eigenspace (geometric multiplicity).
8	"Diagonalizable $\Rightarrow$ n distinct eigenvalues."	False: I_n is diagonalizable with one eigenvalue.
9	Columns of P and entries of D in different orders.	Column i of P must be an eigenvector for the ith entry of D.
10	A^k = raise each entry to the power k.	$A^k = PD^kP^{-1}$; only D^k is done entry by entry.

Common mistake: a full wrong solution, corrected

Question. Eigenvalues of $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

Wrong: "$R_2 \to R_2 - 2R_1$ gives $\begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}$, so $\lambda = 1, -3$."

Right: $\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ -2 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - 4 = (\lambda - 3)(\lambda + 1)$, so $\lambda = 3, -1$. Check: $3 + (-1) = 2 = \operatorname{tr} A$ and $(3)(-1) = -3 = \det A$ ✓.

19 Everything on one page

Definition. $A\mathbf{x} = \lambda\mathbf{x}$, $\mathbf{x} \neq \mathbf{0}$. Eigenvector: does not turn. Eigenvalue: stretch factor.

Eigenvectors. Basis of the null space of $\lambda I - A$ (row reduce, free variables $\to$ vectors).

Invertibility. A invertible $\iff$ 0 is not an eigenvalue.

Diagonalize. n independent eigenvectors $\Rightarrow$ $P = [\mathbf{p}_1 \cdots \mathbf{p}_n]$, $P^{-1}AP = D$.

Test. n distinct $\lambda \Rightarrow$ diagonalizable. Repeated λ: geometric $=$ algebraic for every λ?

Eigenvalues. Solve $\det(\lambda I - A) = 0$. For 2×2: $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$. Check: $\sum\lambda = \operatorname{tr} A$, $\prod\lambda = \det A$.

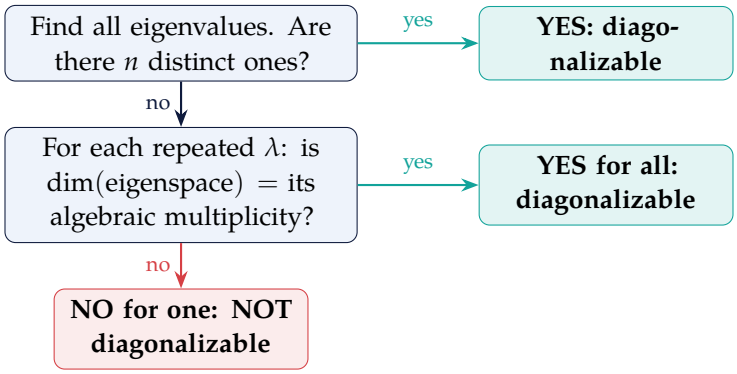
Triangular. Eigenvalues $=$ diagonal entries.

Powers. $A\mathbf{x} = \lambda\mathbf{x} \Rightarrow A^k\mathbf{x} = \lambda^k\mathbf{x}$.

High powers. $A^k = PD^kP^{-1}$.

Multiplicities. $1 \leq$ geometric $\leq$ algebraic. Geometric $= n - \operatorname{rank}(\lambda I - A)$.

The decision tree for ``Is A diagonalizable?''



20 Practice

Solved in the separate file (with every step): problems from Anton Exercise Sets 5.1 and 5.2 and Williams Exercise Set 3.4, with the same numbers as in the books.

Try yourself first: Anton 5.1: 2, 4, 6, 8, 10, 12, 14, 16, 25, 27; Anton 5.2: 2, 4, 6, 8, 10, 12, 14, 16, 18; Williams 3.4: 2, 4, 6, 10, 12, 16.

Thank you

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